ABIGX: A Unified Framework for Explainable Fault Detection and Classification

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***Abstract*—This paper proposes ABIGX (Adversarial fault reconstruction-Based Integrated Gradient eXplanation), a unified framework for explainable fault detection and classification (FDC). ABIGX builds on the foundational principles of established fault diagnosis methods, including contribution plots (CP) and reconstruction-based contribution (RBC), while extending their applicability to general FDC models and improving fault explanation results. Central to ABIGX is the Adversarial Fault Reconstruction (AFR) method, which rethinks fault reconstruction from the perspective of adversarial attacks, introducing a novel fault index applicable to both fault detection and classification tasks. In fault detection, we theoretically bridge ABIGX with conventional fault diagnosis methods by proving that CP and RBC are the linear specifications of ABIGX. For fault classification, we address the challenge of fault class smearing, an inherent issue that can obscure accurate explanations. We demonstrate that ABIGX effectively mitigates this issue, outperforming current gradient-based explanation methods. The experiments evaluate the explanations of FDC by quantitative metrics and intuitive illustrations. The results validate the generality and accuracy of AFR, and show that ABIGX provides more comprehensive and precise explanations across various FDC models, offering a significant improvement over existing methods.**

***Index Terms*—Feature attribution, integrated gradient, eXplainable artificial intelligence (XAI), counterfactual explanation.**

I. Introduction

DATA-DRIVEN fault detection and classification (FDC) is an important technique for recognizing patterns of faults in industrial processes. Fault detection [1] decides whether a fault/anomaly has occurred, which is realized by measuring the residual projection of unsupervised models like Principal Component Analysis (PCA) [2] and AutoEncoder (AE) [3]. Fault classification involves identifying the types of fault/anomaly, where the artificial neural networks (NN) [4] have achieved great success recently.

Nevertheless, these complex models become increasingly opaque, tending to be "black boxes" for the users. This opacity leads to the lack of trust in the model output (predictions), which is vital for practical applications in critical fields (e.g., industrial processes, autonomous systems and healthcare). Explainable AI (XAI) [5] is to increase the transparency and trust of AI models, by seeking explanations for how predictions come about.

Within the field of XAI, eXplainable FDC (XFDC) aims to help people understand the underlying factors that contribute to faults. Given a fault sample, XFDC indicates how much each variable contributes to the monitoring results (detection score or classification confidence). The variables with high contributions are considered as the potential root cause of the anomaly (e.g., faulty sensors). By identifying these variable contributions, XFDC can support users in explaining the faults predicted by models and proactively repairing defective components.

Contribution Plot (CP) [6] and Reconstruction-Based Contribution (RBC) [7] are foundational in fault diagnosis, offering intuitive variable-level attributions based on residual decomposition and reconstruction error, respectively. CP highlights fault-relevant variables via SPE decomposition, while RBC links deviations to model reconstruction. These methods provide valuable diagnostic insights but primarily effective for linear PCA-based fault detection models and assume variable independence.

As for the extension to nonlinear models, while CP can be heuristically extended to nonlinear models, it lacks a principled foundation in nonlinear contexts. RBC, by contrast, is tightly coupled with linear reconstruction and requires substantial modifications to handle nonlinearity [8], [9], [10], [11], [12]. Moreover, these fault diagnosis[1] extensions are generally constrained to specific structures (e.g., KernalPCA and Stacked AE) and are only applicable to fault detection tasks. This limitation creates a significant gap in general-purpose fault explanation across modern deep learning-based FDC systems.

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The implementation code is released in the public repository: https://github.com/qianjc3391/ABIGX.

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[1] In previous works, diagnosing the cause of a fault by contributing variable significance to the fault detection indices is often referred to as *fault diagnosis* [7] or *fault isolation* [13].

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In this work, we propose a unified framework called ABIGX (Adversarial Fault Reconstruction-Based Integrated Gradient eXplanation). ABIGX builds on the conceptual strengths of CP and RBC (e.g., residual- and reconstruction-based reasoning). ABIGX first rethinks fault reconstruction (FR) [14] from the perspective of adversarial attacks [15], and then calculates the variable contributions with deep learning attribution method, Integrated Gradients (IG) [16]. Our framework is built on two key motivations:

1) Extending fault explanation to general FDC models in one framework: While traditional explainers like CP and RBC offer conceptually sound approaches with solid theoretical foundation, they are inherently constrained to linear models and lack generalizability to modern, nonlinear FDC systems. Our method, ABIGX, is built upon the core principles of CP and RBC, but reinterprets them through the adversarial optimization. Specifically, we introduce Adversarial Fault Reconstruction (AFR), which uses adversarial attack techniques to generate semantically valid counterfactuals that suppress the fault signal. When combined with Integrated Gradients (IG), this enables ABIGX to generalize fault attribution to any differentiable FDC model, including PCA, AE, MLP, CNN, and Vision Transformers (ViT). By retaining the diagnostic intuition of CP and RBC while overcoming their architectural limitations, ABIGX offers a unified and broadly applicable framework that advances fault explainability in both linear and nonlinear domains.

2) Improving fault explanation accuracy with theoretical foundation: Another major limitation of traditional explainers is their tendency to treat variables independently, overlooking critical interactions between variables—especially in complex, high-dimensional, nonlinear models. By adopting the adversarial attack perspective, our method automatically accounts for these variable interconnections, significantly improving the accuracy and comprehensiveness of fault explanations. Built upon and extending the conceptual foundation of CP and RBC, the ABIGX framework provides a robust theoretical foundation for generating more precise explanations, as demonstrated in the following theoretical analyses and experimental results.

By addressing these two key motivations, our framework, ABIGX, not only enhances the generalization of fault explanations across various models but also significantly improves their accuracy. Fig 1 illustrates the relationships between the major methods involved in the paper, showing the generality of ABIGX. In summary, the contributions of this work are four-fold:

- We rethink the fault reconstruction from the adversarial attack perspective and propose AFR for the general FDC models. The new SPE index extends AFR to the reliable fault reconstruction for classification, which was less considered before. (Section III)
- We propose a unified framework, ABIGX, for effectively explaining general FDC models. We demonstrate that ABIGX includes the canonical fault diagnosis methods in a bigger framework, by proving CP and RBC are the linear specifications of ABIGX (Section IV and Section VI).
- We introduce a new problem in explainable fault classification, known as fault class smearing, which is an inherent issue in fault classification models that leads to misleading variable contributions and inaccurate model explanations. Through theoretical analysis, we prove that ABIGX effectively mitigates this problem, resulting in its superior explainability for fault classification models (Section V).
- We perform experiments on both sensor (Tennessee Eastman Process) and image (wafer map fault, MVTec anomaly detection) datasets with multiple quantitative metrics. Our evaluation spans a diverse range of models, including classical PCA, AE, MLP, CNN, and ViT models, demonstrating that ABIGX consistently produces superior explanations compared to existing XAI methods (Section VII).

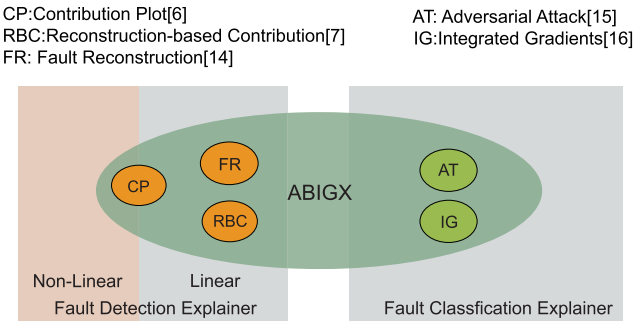


Fig. 1. The illustration of ABIGX and related methods. ABIGX is a explanation framework applicable for both linear and non-linear fault detection and classification models. It frames two classical fault diagnosis methods (CP and RBC) and generalizes to general FDC models aided by adversarial attack and IG method.

The remainder of the paper is organized as: Section II introduces the preliminaries and related works in the fields of FDC, fault diagnosis, adversarial attack, and XAI; Section III proposes AFR by rethinking fault reconstruction from adversarial attack; Section IV presents the unified framework, ABIGX, and proves its explainability by analyzing the fault class smearing problem; Section VI theoretically bridges the canonical fault diagnosis methods with ABIGX; Section VII reports the experiments for evaluating our proposal; Section VIII concludes the paper.

II. Preliminaries and Related Works

Table I lists the descriptions of major symbols in this paper. The less frequent symbols not listed in the table are defined at their first occurrence within the text.

A. FDC

Fault detection aims to model normal operating conditions using statistical or learning-based methods. These models typically project data into two subspaces: a principal component subspace capturing normal patterns, and a residual subspace representing noise or anomalies. A fault is detected when the sample's residual exceeds a predefined *control limit*. Traditional methods are based on PCA [17], [18], [19], while more recent approaches use autoencoders (AEs) [20], [21], [22].

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TABLE I
NOTATION TABLE FOR MAJOR SYMBOLS IN PAPER

Symbols	Descriptions
$\mathbf{x}$	Input sample vector in FDC models.
$\mathbf{P}$	Principal loading vector in PCA.
$\tilde{\mathbf{C}}$	Residual projection matrix, $\tilde{\mathbf{C}} = \mathbf{I} - \mathbf{P}\mathbf{P}^T$
SPE	Squared prediction error
$\mathbf{F}_{\text{model}}$	The FDC models, e.g., PCA, AE, NN
ξ_i, f_i	Identity vector and scalar indicating fault on the i-th variable with magnitude f_i
$\mathbf{x}'$	Adversarial sample for $\mathbf{x}$
$\mathbf{x}'_{AFR}$	Adversarial fault reconstructed sample for $\mathbf{x}$, simultaneously in all variables
$\mathbf{x}'_{AFR^{OV}}$	Adversarial fault reconstructed sample for $\mathbf{x}$, one variable at a time
explainer_i	i-th variable contribution by explainers, e.g., CP, RBC, Grad, IG, ABIGX
$FCS_{\text{explainer}}$	Fault class smearing degree of explainer

Given a fault-free dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_1, \ldots, \mathbf{x}_m\} \in \mathbb{R}^{m\times n}$, fault detection models learn to minimize reconstruction or residual error. The model output is denoted as $\mathbf{F}_{FD}(\mathbf{x}) \in \mathbb{R}^{m\times n}$, referred to as the *principal component sample*.[2]

In this work, we focus on encoder-decoder-based detection models. The nonlinear case uses AEs with neural network encoders and decoders:

$$\mathbf{F}^{AE}(\mathbf{x}) = \mathbf{F}_{dec}(\mathbf{F}_{enc}(\mathbf{x})) \tag{1}$$

PCA can be viewed as a linear AE:

$$\mathbf{F}^{PCA}(\mathbf{x}) = \mathbf{P}\mathbf{P}^T\mathbf{x} \tag{2}$$

where $\mathbf{P} \in \mathbb{R}^{n\times l}$ is the loading matrix, projecting $\mathbf{x}$ onto the top l principal components via eigendecomposition of the data covariance.

SPE index for fault detection in this work is based on squared prediction error (SPE), which is the ℓ_2 norm measurement of residual error:

$$SPE(\mathbf{x}; \mathbf{F}_{FD}) = \|\mathbf{x} - \mathbf{F}_{FD}(\mathbf{x})\|_2^2 \tag{3}$$

The control limit δ^2 is derived by the confidence limits of SPE values over normal data distribution. The sample is predicted as normal when its SPE is lower than the control limit.

Fault classification is to recognize the different patterns in the fault samples. Unlike fault diagnosis, classification is a supervised task with labeled training datasets. This work focuses on the most popular classification model: neural networks.

Given a dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_1, \ldots, \mathbf{x}_m\} \in \mathbb{R}^{m\times n}$ and label $Y \in \{0, 1, \ldots, k\}$ with k being the number of fault types. The fault classifier $\mathbf{F}_{FC} : \mathbb{R}^n \to \{0, 1, \ldots, k\}$ is trained by:

$$\mathbf{F}_{FC} = \arg\min_{\mathbf{F}} \mathbb{E}_{\mathbf{x}\in\mathbf{X},\ y\in Y} J(y, \mathbf{F}(\mathbf{x})) \tag{4}$$

where J is the cross-entropy loss function for NN.

[2]To avoid confusion with fault reconstruction, we use the term "principal component sample" to refer to outputs of both PCA and AE models.

B. Fault Diagnosis

The explanation of fault detection models—often referred to as fault diagnosis—has been extensively studied [6], [7], [9], [23]. Once a fault is detected, fault diagnosis aims to identify its root variable by quantifying how much each input variable contributes to fault indicators such as the Squared Prediction Error (SPE). Below, we briefly review three classical fault diagnosis methods commonly used with PCA-based models.

Fault reconstruction [14] seeks the minimal adjustment needed to return a fault sample's indicator (e.g., SPE) to a normal range. This adjustment is defined by two parameters: the direction ξ_i (a unit vector representing the i-th variable) and a scalar magnitude f_i. For example, with five sensor variables, the direction vector for the first variable is:

$$\xi_i = (1 \quad 0 \quad 0 \quad 0 \quad 0)^T \tag{5}$$

Most previous fault reconstruction methods predefine the reconstruction variable ξ_i in turn and solve the magnitude f_i. Given ξ_i, the fault reconstruction is to find f_i:

$$f_i = \arg\min SPE(\mathbf{x} - f_i\xi_i; \mathbf{F}_{FD}) \tag{6}$$

where $\mathbf{x}$ is the fault sample and $\mathbf{x} - f_i\xi_i$ is the reconstructed sample brought back to the normality.

The fault direction can be determined by the direction ξ_i such that the SPE of the reconstructed sample is minimized. However, fault reconstruction cannot provide accurate contributions for each variable, which is solved by RBC.

RBC [7] is to calculate the fault index of the reconstruction vector $f_i\xi_i$, which is expressed as:

$$\begin{aligned} RBC_i \equiv SPE(f_i\xi_i; \mathbf{F}_{PCA}) &= \|f_i\xi_i - \mathbf{P}\mathbf{P}^T f_i\xi_i\|_2^2 \\ &= (f_i\xi_i)^T(\mathbf{I} - \mathbf{P}\mathbf{P}^T)f_i\xi_i \\ &= (f_i\xi_i)^T\tilde{\mathbf{C}}f_i\xi_i \end{aligned} \tag{7}$$

where $\tilde{\mathbf{C}} = \mathbf{I} - \mathbf{P}\mathbf{P}^T$ is the projection matrix to residual subspace.

CP [6] is directly constructed by determining the contribution of each variable to the detection index. Firstly, SPE defined can be expanded:

$$SPE(\mathbf{x}; \mathbf{F}_{FD}) = \Sigma_{i=1}^n(\mathbf{x}_i - \mathbf{F}_{FD}(\mathbf{x})_i)^2 \tag{8}$$

where $\mathbf{x}_i$ and $\mathbf{F}_{FD}(\mathbf{x})_i$ are the i-th variable of detected sample and principal component sample. The CP contribution on SPE, CP_i, can be simply calculated by the squared value difference between the i-th variable of the input and output sample:

$$\begin{aligned} CP_i &= (\mathbf{x}_i - \mathbf{F}_{FD}(\mathbf{x})_i)^2 \\ &= (\xi_i^T\tilde{\mathbf{C}}\mathbf{x})^2 \qquad \text{for PCA} \end{aligned} \tag{9}$$

C. Gradient-Based XAI Methods

Gradient-based XAI method can directly attribute the input feature through backward propagation, which are computationally efficient and applicable for multiple model modalities. Recently, Integrated Gradient (IG) [16] is a popular gradient-based method and shares some identical intrinsic properties with fault

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diagnosis methods. Hence, this paper mainly focuses on IG and another canonical method: saliency map (naive gradient) [24].

Saliency map directly provides the explanation by computing the gradient of output prediction w.r.t. the input variables, which is a natural analogue of model behavior:

$$Grad_i = \frac{\partial \mathbf{F}(\mathbf{x})}{\partial x_i} \tag{10}$$

where the SPE index and logit value of the ground-truth class are used as the model $\mathbf{F}$ output for fault detection and classification, respectively.

IG is a popular XAI algorithm. Instead of simply computing the gradient, IG accumulates the gradients along a path $\gamma(\alpha)$ ($\alpha \in [0,1]$) from the explained sample $\mathbf{x}$ to a counterfactual baseline $\mathbf{x}'$.[3] The IG contribution of i-th variable is defined as:

$$IG_i = \int_{\alpha=0}^{1} \frac{\partial \mathbf{F}(\gamma(\alpha))}{\partial \gamma_i(\alpha)} \frac{\partial \gamma_i(\alpha)}{\partial \alpha} \, d\alpha \tag{11}$$

where $\gamma(0) = \mathbf{x}'$ is the baseline, $\gamma(1) = \mathbf{x}$ is the explained fault sample, and the f is the same FDC output function as defined in the vanilla gradient.

D. Adversarial Attack

Adversarial attack is a critical security issue of ML models, adversarial robustness of FDC models has been studied in the work [25]. Concerning the attacks on fault samples, the adversarial examples $\mathbf{x}'$ is solved by:

$$\begin{aligned} \mathbf{x}' = \arg\min \mathbf{F}(\mathbf{x}') \\ \text{s.t. } D(\mathbf{x}, \mathbf{x}') < \eta, \\ y' \neq y \end{aligned} \tag{12}$$

where $\mathbf{F}$ is the same as defined in (10), D is the distance measurement, η is a predefined distance constraint, y is the fault label and y' is the predicted label of $\mathbf{x}'$ (the second constraint is not strictly necessary).

E. Challenges in Explainable FDC

Generating accurate and theoretically grounded explanations for deep neural network-based FDC models remains a key challenge in explainable fault detection and classification (FDC), despite growing research interest. While both traditional fault diagnosis methods and deep learning explainers have been applied, several limitations persist:

Challenges in fault diagnosis: Classical fault diagnosis methods, such as CP and RBC, are well-studied for linear models. Although some extensions to nonlinear models have been proposed [8], [9], [10], [11], [26], they often treat each variable independently and overlook important interactions—especially in complex nonlinear systems. Moreover, most of these methods are limited to fault detection and cannot be directly applied to fault classification tasks.

[3]The counterfactual samples contrastively explain the FDC models by answering:"Which features cause the model to predict fault rather than normal?" The normal sample is counterfactual to the explained anomaly sample.

Challenges in deep learning explanation: Gradient-based explainers like Integrated Gradients (IG) have shown promise in interpreting deep models, but applying them to FDC introduces new difficulties. A major issue is the choice of baseline: common baselines such as zero or constant vectors [27] are often semantically meaningless in FDC settings, where distinguishing faults from normal conditions is critical. In addition, many deep learning explanation methods lack a strong theoretical foundation specific to the fault diagnosis domain.

Our proposed framework, ABIGX, addresses these gaps by combining the strengths of traditional fault diagnosis and deep learning explanation. Rooted in the principles of CP and RBC, ABIGX introduces adaptive, fault-aware baselines into IG, enabling accurate, model-agnostic explanations for both fault detection and classification tasks.

III. Fault Reconstruction: Adversarial Perspective

First of all, we study the fault reconstruction from the adversarial attack perspective and state that the existing fault reconstruction solves the same optimization problem with the sparse adversarial attack.

Theorem 1: For fault detection, the existing fault reconstruction shares the same optimization objective with the sparse adversarial attack on fault samples.

Proof: According to (12), the sparse adversarial attack on fault detection can be defined with the ℓ_0 norm constrain [4]:

$$\begin{aligned} \mathbf{x}' = \arg\min SPE(\mathbf{x}'; \mathbf{F}_{FD}) \\ s.t. \ \|\mathbf{x}', \mathbf{x}\|_0 < \eta \end{aligned} \tag{13}$$

where $\|\cdot\|_0$ measures the non-zero element number of the vector and $\eta \in \mathbb{N}^+$ controls the variable number to be attacked.

If $\eta = 1$ for attacking only one variable and the attacked variable direction ξ_i are given, the attack objective becomes computing the magnitude f_i on this direction with $\mathbf{x}' = \mathbf{x} + f_i\xi_i$:

$$f_i = \arg\min SPE(\mathbf{x} - f_i\xi_i; \mathbf{F}_{FD}), \ where \ \|\xi_i\|_0 = 1 \tag{14}$$

which is the same as fault reconstruction defined in (6). □

A. Adversarial Fault Reconstruction

Theorem 1 inspires us to use the adversarial attack algorithms for more generalized fault reconstruction, where the one-by-one predefining reconstruction direction is unnecessary. Iteratively predefining direction and solving magnitude could be time-consuming and ignore the interaction between variables.

Hence, we propose Adversarial Fault Reconstruction (AFR) for FDC models. Instead of searching the optimal fault directions one-by-one with brute force, AFR can automatically solve the

[4]The fault detection relies on the SPE of model reconstruction (3) to predict faults: normal samples are trained to yield low reconstruction errors, while fault samples exhibit higher values. To adversarially attack fault samples to be predicted as the normal, the adversarial samples must exhibit low SPE values for model reconstructions (being lower than control limit is not strictly required here).

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(combination of) variables minimizing the SPE indices by applying the magnitude constraint on the distance between explained samples $\mathbf{x}$ and reconstructed samples $\mathbf{x}'_{AFR}$.

Definition 1 (AFR for FDC): AFR reconstructs the fault in FDC models with the following optimization problem:

$$\mathbf{x}'_{AFR} = \arg\min_{\mathbf{x}'} SPE(\mathbf{x}'; \mathbf{F}_{FDC}) \quad s.t. \ \|\mathbf{x} - \mathbf{x}'\|_p \leq \eta \tag{15}$$

η is to constrain the distance between reconstructed and original fault samples, the measurement of which is ℓ_p norm ($p = 0, 1, 2$).

Both constrained ℓ_1 and ℓ_2 norm magnitude could enable the attack algorithm automatically to focus on the variables that contribute most to the SPE index. AFR is more general since the variable interaction is more complex in the nonlinear models, and one-by-one variable search is more computationally expensive in the high-dimensional samples like images.

Notably, we declare that the conventional one-by-one variable fault reconstruction can be regarded as a specification of AFR, which is named as AFR-OneVar:

Definition 2 (AFR-OneVar): AFR-OneVar computes a set of reconstructed samples $\{\mathbf{x}'^{(1)}_{AFR^{OV}}, \mathbf{x}'^{(2)}_{AFR^{OV}}, \ldots, \mathbf{x}'^{(n)}_{AFR^{OV}}\}$ respectively for n variables, where n optimization problems are involved.

Given i-th variable direction ξ_i, the $\mathbf{x}'^{(i)}_{AFR^{OV}}$ can be defined as (14):

$$\mathbf{x}'^{(i)}_{AFR^{OV}} = \arg\min_{\mathbf{x}'} SPE(\mathbf{x}'; \mathbf{F}_{FDC}) \quad s.t. \ \mathbf{x}' - \mathbf{x} = f_i \xi_i \tag{16}$$

where $i \in \{0, 1, \ldots, n\}$.

This is similar to some previous fault diagnosis methods [26] that extend RBC to nonlinear AE models. Fig. 5 intuitively illustrates the comparison of AFR and AFR-OneVar with a 2-D PCA detection model.

Previously, the SPE in (15) is only defined for fault detection. To make AFR also applicable for fault classification, we introduce a novel SPE index for the classification version.

B. SPE for Fault Classification

The SPE for fault classification is derived from fault detection (defined in (3)). Whereas the detection models are trained to minimize the residual projection of normal samples, we argue that the barycenter of normality samples' residual projection is on the zero point of residual subspace. Hence, the detection SPE for fault samples can be regarded as computing the Euclidean distance in residual subspace between fault samples' projection and barycenter of normality samples' projection (i.e., zero point).

Thus, SPE of fault classification can be defined similarly. We compute the Euclidean distance in the representation space.[5] The barycenter of normality samples' representation is the "zero point" and the SPE is defined as the Euclidean distance between the normality barycenter and fault representation.

[5]Commonly, the penultimate layer's output of classification neural network is chosen as the representation.

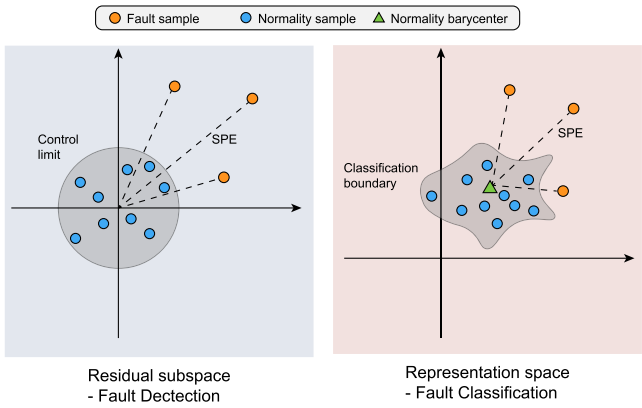


Fig. 2. SPE for fault classification (right), compared with the fault detection SPE (left).

Fig. 2 illustrates the comparison with detection SPE and classification SPE. The definition of classification SPE is:

Definition 3 (SPE for fault classification): Given the representation layer of classification network $\mathbf{F}_h$, the set of normality samples $\mathbf{X}_{nor}$, the classification SPE of fault sample $\mathbf{x}$ is:

$$SPE(\mathbf{x}; \mathbf{F}_{FC}) = \|h(\mathbf{x}) - \mathbb{E}_{\mathbf{x}_{nor} \sim \mathbf{X}_{nor}}[\mathbf{F}_h(\mathbf{x}_{nor})]\|_2^2 \tag{17}$$

where the term $\mathbb{E}_{\mathbf{x}_{nor} \sim \mathbf{X}_{orn}}[\mathbf{F}_h(\mathbf{x}_{nor})]$ is the barycenter of normality representations.

With defined SPE indices, AFR for FDC in (15) can be solved with adversarial attack algorithms, which are briefly discussed in the following section.

C. AFR Algorithms

For different FDC models, the objective of AFR (15) can be solved by different optimization algorithms. We mainly consider two specific algorithms: the mathematical programming that can guarantee the optimal solution and the gradient descent that is more general for multiple models.

1) AFR via Mathematical Programming: For the linear models, we can convert AFR into a convex mixed-integer linear programming (MILP), where the global minima can be solved. Here we specify the problem of ℓ_0 norm constraint, which is appealing since it can directly diagnose the most η significant variables in the FDC models.

Firstly, a set of auxiliary binary variables ρ is introduced to formulate the ℓ_0 norm constraint into the mixed-integer linear form:

$$\|f\xi\|_0 < \eta \equiv \Bigg[\left(\sum_j \rho_j < \eta \right) \wedge (\mathbf{x}'_j \leq \mathbf{x}_j + \rho_j \cdot \epsilon) \wedge (\mathbf{x}'_j \geq \mathbf{x}_j - \rho_j \cdot \epsilon) \wedge (\rho_j \in \{0, 1\}) \Bigg] \tag{18}$$

where conjunction symbol $\wedge$ denotes logical AND, $\mathbf{x}'_j$ denotes the j-th variable of the reconstructed sample, and ϵ is a large predefined bound for all variables.

Next, the objective $SPE(\mathbf{x}'; \mathbf{F}_{FDC})$ of linear models is formulated by:

$$\mathbf{x}'^T \tilde{C} \mathbf{x}' \tag{19}$$

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where $\tilde{C}$ is the residual projection matrix.

This idea to solve AFR by MILP is similar to adversarial verification works [28], [29], [30], which can be extended to ReLU-based NN. However, without loss of generality, this work applies the gradient-based attack algorithms for solving AFR, which is discussed in the following.

2) AFR via Gradient Descent: For more general FDC models, we can realize AFR by utilizing the existing attack algorithms. In particular, we apply a multi-step gradient attack, Projected Gradient Descent (PGD) [31], which is applicable for both ℓ_0, ℓ_1 and ℓ_2 norm:

$$\mathbf{x}'_{t+1} = \mathbf{x}'_t - \frac{\nabla_{\mathbf{x}_t} SPE(\mathbf{x}_t; \mathbf{F}_{FDC})}{L} \tag{20}$$

where $\mathbf{x}'_t$ is the adversarial sample at t-th iteration and L is to normalize the perturbation under certain norm.[6]

D. AFR v.s. Adversarial Attack

Although AFR can be viewed as an adversarial attack on fault samples, it has unique characteristics that set it apart. AFR requires the attacked fault sample to mimic the model presentation of normal samples, whereas conventional adversarial attacks typically focus on causing the model to misclassify the adversarial samples. Consequently, the objective of AFR differs from standard adversarial attacks. Specifically, AFR minimizes the model representation distance between the adversarial fault sample and the normal sample (measured by the SPE), while conventional adversarial attacks simply minimize the model prediction of the adversarial samples.

Because AFR shares similarities with adversarial attacks in its objective, it can be directly solved using a variety of attack algorithms rather than being restricted to a specific type. In this paper, we primarily utilize the PGD algorithm, a typical and standard approach, applicable in general FDC models.

IV. ABIGX: A Unified Framework

Aided by adversarial attack algorithms and the new classification SPE, AFR generalizes the fault reconstruction to general FDC models. Based on the counterfactual samples reconstructed by AFR, we are dedicated to computing the variables' contributions to the model prediction.

Recalling the idea of RBC, RBC is to calculate the amount of model output (fault index) along the reconstruction vector $f_i\xi_i$. This implies the same motivation as the IG [16] methods, which also computes the contribution amount to the model output along a path. The concrete relationship between RBC and IG will be rigorously proved in Section VI.

Beyond the linear RBC and FR, IG and AFR correspondingly extend the variable contribution and fault reconstruction to general model modalities for both fault detection and classification tasks. Hence, by introducing the IG method into AFR, we propose ABIGX, which is a unified framework for explaining general FDC models.

[6]If not specified, AFR and $ABIGX$ use the ℓ_2 norm as the default distance constraint.

A. Definition

In general, ABIGX integrates gradients of model output along the path from the explained sample to the AFR-reconstructed sample.

Definition 4 (ABIGX): ABIGX explains FDC models by attributing the variables: Given the explained sample $\mathbf{x}$ and reconstructed sample $\mathbf{x}'_{AFR}$ by AFR, the ABIGX contribution of i-th variable is defined as:

$$ABIGX_i = (\mathbf{x}_i - \mathbf{x}'_{AFRi}) \times \int_{\alpha=0}^{1} \frac{\partial \mathbf{F}_{FDC}(\mathbf{x}'_{AFR} + \alpha(\mathbf{x} - \mathbf{x}'_{AFR}))}{\partial \mathbf{x}_i} d\alpha \tag{21}$$

Equation (21) is a Riemann sum of (11), where the gradient is integrated along the straight-line path from $\mathbf{x}'_{AFR}$ to $\mathbf{x}$. If the reconstructed samples are solved in the one-by-one direction, the contribution can also be calculated along the direction one-by-one, which is named as ABIGX-OneVar:

Definition 5 (ABIGX-OneVar): ABIGX-OneVar contribution of i-th variable is to integrate gradients along the path from $\mathbf{x}$ to the AFR-reconstructed sample $\mathbf{x}'^{(i)}_{AFR^{OV}}$ on variable direction ξ_i:

$$\begin{aligned} ABIGX_i^{OV} &= (\mathbf{x}_i - \mathbf{x}'^{(i)}_{AFR^{OV}}) \\ &\quad \times \int_{\alpha=0}^{1} \frac{\partial \mathbf{F}_{FDC}(\mathbf{x}'^{(i)}_{AFR^{OV}} + \alpha(\mathbf{x} - \mathbf{x}'^{(i)}_{AFR^{OV}}))}{\partial \mathbf{x}_i} d\alpha \\ &= f_i\xi_i \times \int_{\alpha=0}^{1} \frac{\partial \mathbf{F}_{FDC}(\mathbf{x} + (1-\alpha) f_i\xi_i)}{\partial \mathbf{x}_i} d\alpha \end{aligned} \tag{22}$$

ABIGX-OneVar omits the variables' interaction and can be regarded as the extension of RBC for general FDC models. Since both AFR and IG only require the gradient information of the model, ABIGX is a general explanation technique for FDC models.

B. Algorithm

Algorithm 1 presents the ABIGX attribution method, which combines Adversarial Fault Reconstruction (AFR) and Integrated Gradients (IG) to generate faithful feature attributions for explaining FDC models. The method consists of two stages. In Stage I, AFR reconstructs a fault-suppressing baseline by moving the input toward a model-induced normality region in the chosen representation space, operationalized by minimizing the SPE distance to the normal-representation barycenter under an ℓ_p constraint. We emphasize that this is an approximate projection defined by the model and the constraint set, rather than a guarantee of landing exactly on the true data manifold. In Stage II, IG algorithm is applied along a linear path from the AFR-reconstructed baseline to the fault input, replacing the conventional baselines (e.g., zero sample) that used in most IG-based method.

For ABIGX-OneVar, Algorithm 1 is repeated for each variable individually, and the final attribution is derived by merging all the variable attribution together.

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Algorithm 1: ABIGX Attribution for FDC Models.

Input: Explained fault sample $\mathbf{x} \in \mathbb{R}^n$,
Normality samples $\{\mathbf{x}_{nor}^i\}_{i=1}^N \in \mathbb{R}^n$,
FDC model $\mathbf{F}$, representation layer $\mathbf{F}_h$
Params: Steps T, step size η
Output: Feature attribution $A \in \mathbb{R}^n$

Step 1: Adversarial Fault Reconstruction (AFR)
// Initialize normality representation and AFR
$h_{nor} \leftarrow \mathbb{E}_{\mathbf{x}_{nor} \sim \mathbf{X}_{nor}}[\mathbf{F}_h(\mathbf{x}_{nor})]$ for FC, and 0 for FD
$\mathbf{x}'_{AFR} \leftarrow \mathbf{x}$
// Compute AFR, Definition 1 and Eq. 20
for $t = 1$ **to** T **do**
  $SPE = \|\mathbf{F}_h(\mathbf{x}'_{AFR}) - h_{nor}\|_2^2$
  $g_t \leftarrow \nabla_{\mathbf{x}_t} SPE(\mathbf{x}_t; \mathbf{F})$ // Grad. of SPE
  $\mathbf{x}'_{AFR} \leftarrow \mathbf{x}'_{AFR} + \eta \cdot g_t / |g_t\|_2^2$ // Update

Step 2: IG-based Attribution (ABIGX)
Compute difference: $\Delta\mathbf{x} \leftarrow \mathbf{x} - \mathbf{x}'_{AFR}$
Initialize total attribution: $A \leftarrow 0$
// Integrate Gradients towards AFR, Definition 4
foreach $\alpha \in linspace(0, 1, T)$ **do**
  $\mathbf{x}_\alpha \leftarrow \mathbf{x}'_{AFR} + \alpha \cdot \Delta x$
  $g_\alpha \leftarrow \nabla_{\mathbf{x}} \mathbf{F}(x_\alpha)$ // Grad. of FDC output
  $A \leftarrow A + g_\alpha$ // Update attribution
Finalize attribution: $A \leftarrow \frac{1}{T} A \cdot \Delta x$
return A *as* $ABIGX$

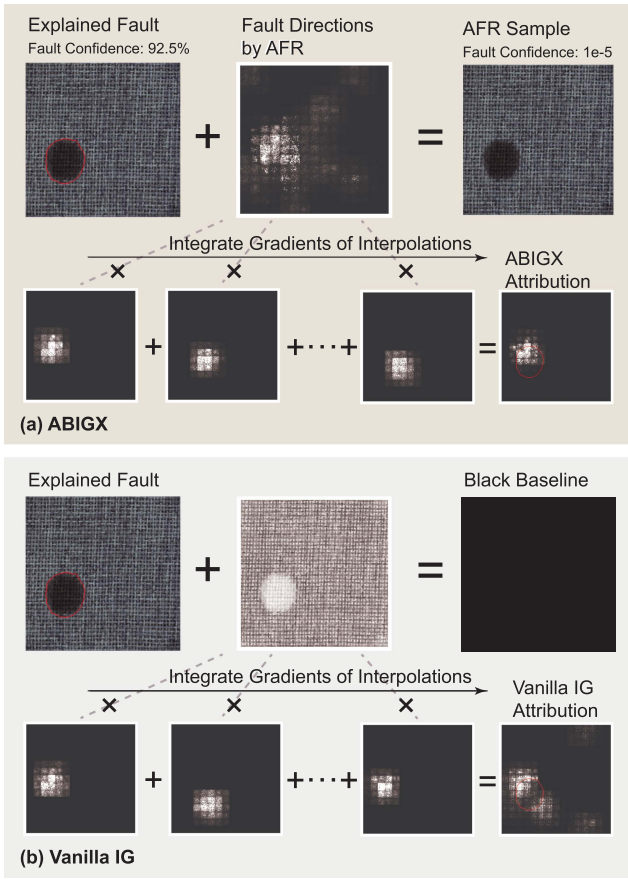


Fig. 3. Comparison of ABIGX (a) and Vanilla IG (b) explanations for the WinClip [32] model using an anomaly image from MVTec-AD [33]. ABIGX differs by using AFR to adaptively find a semantically meaningful, fault-suppressing baseline, leading to more informative path gradient integration.

C. Discussion

1) Anomaly Image Explanation Example: The ABIGX framework extends Integrated Gradients (IG) by introducing an adaptive baseline selection through Adversarial Fault Reconstruction (AFR). As illustrated in Fig. 3, given an anomaly image, ABIGX searches for a AFR baseline with low fault confidence by minimizing the classification SPE (Definition1). Although the perturbation from the original sample is visually negligible, it significantly alters the model's prediction, effectively projecting the fault input onto the model-induced normality region. This AFR baseline is semantically meaningful and aligned with the model's internal representation. The vector from this baseline to the original fault sample captures the essential fault-inducing perturbations, forming a more informative integration path for gradient accumulation.

Once the AFR baseline is obtained, ABIGX proceeds to compute feature attributions by a simple IG method along a straight-line path from the AFR baseline to the original sample, as shown in the figure. This process contrasts with vanilla IG, which uses a fixed, semantically-agnostic black image as the baseline. By the fault-aware integration path, ABIGX produces more focused and interpretable attribution maps.

2) Tabular Fault Explanations Example: Besides the comparison between different reconstructed fault samples, we give a deep insight into the distinctions between ABIGX and IG, to explain why our proposal is superior for the XFDC.

The essential distinction between ABIGX and IG is the baseline (i.e., the start point of the integral interval): the baseline of ABIGX is the AFR-reconstructed sample, and IG's baseline is the normality sample. We discuss the effect of these two baselines on the variable contribution results. The explanations of TEP Fault 6 in Fig. 6(b) show that the major advantage of ABIGX to IG is Variable XMV(3). Hence, Fig. 4 gives a deep insight into the reason that causes different contributions of XMV(3).

We argue that the two different baselines lead to different paths. The path of IG is more likely to go through the area where other fault types (i.e., fault types other than explained fault and normality) have high confidence, which we call noisy area. The red boxed in Fig. 4 present this area. ABIGX is more likely to avoid the noisy area due to its baseline by AFR, which reconstructs faults on the direction that fastest minimizes the classification SPE. Hence, we suppose that the path of ABIGX is more related to the classification areas of normality and explained fault (e.g., Fault 6). On the contrary, the baseline of the random normality sample may lead IG to the noisy area, the possible case of which is depicted in the top of Fig. 4.

To validate our analysis, Fig. 4 also reports the case of a TEP sample with Fault 6. Firstly, the middle figure reports the

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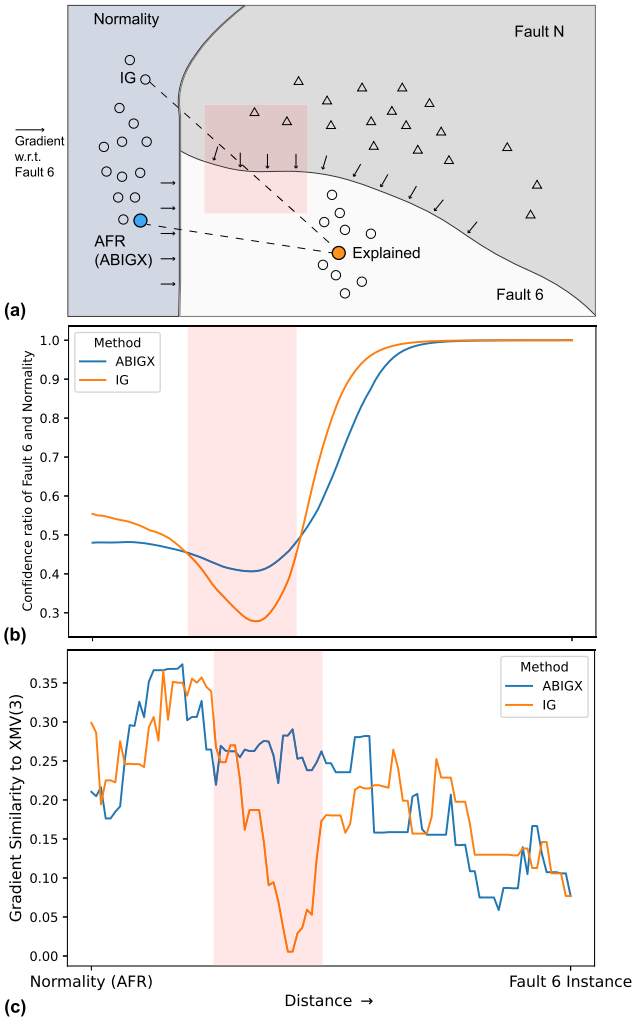


Fig. 4. Comparison of ABIGX and Vanilla IG explanations for TEP [18] fault classification model. (a) The intuitive illustration (not the real cases) of two baselines and integral paths on model boundary; (b) The model confidences along the two paths from two baselines to explained sample; (c) The cosine similarity between gradient direction and the root variable XMV(3), along the same paths. The red boxes in three figures are noisy areas corresponding to each other.

confidence ratio of (the sum of) Fault 6 and normality. When moving from the baseline to explained sample, IG goes through an area where the confidence ratio drops rapidly (red box), which means the confidence of other faults increases largely (e.g., Fault N in the top of Fig. 4). Secondly, the bottom of Fig. 4 reports the gradient similarity to the root variable XMV(3) along the paths. The noisy area gradient is completely irrelevant to XMV(3). We can find that these two areas are heavily overlapped.

Unlike detection, fault classification models the distinction between normality and fault, and also distinguishes different fault types. However, XFDC explains what variables cause the model to predict the fault sample as a "fault" rather than normality. As for why Fault 6 is not predicted as the "Fault N", this is not considered by XFDC. Hence, this is why the noisy area leads to incorrect gradient direction (e.g., the gradient directions in the top of Fig. 4).

Incidentally, the gradient plot in the bottom of Fig. 4 explains why the saliency map is inaccurate: the saliency map only computes the gradient on the explained sample, but the gradient of XMV(3) is much higher when closing to the normality (or AFR-reconstruction).

3) Insights: The core insight behind ABIGX lies in its adaptive utilization of the FDC model's internal representation to construct semantically meaningful baselines for attribution. Unlike vanilla IG approaches that rely on static baselines, such as black images or zero vectors, ABIGX employs AFR to search for a minimally perturbed version of the fault input that significantly reduces the model's fault confidence. This is achieved by minimizing the SPE within the FDC model, effectively projecting the fault sample back onto the model-induced normality region. By doing so, ABIGX ensures that the interpolation path used for gradient integration lies entirely within a semantically valid region of the input space. As illustrated in Figs. 3 and 4, this strategy produces clearer, more focused attribution maps that avoid spurious or noisy gradients typically encountered along unrealistic paths.

Compared to Expected IG, which uses multiple normal samples as baselines, ABIGX offers a more targeted and model-specific baseline construction. While Expected IG improves over vanilla IG by leveraging data-driven baselines, it still lacks adaptiveness to the specific fault instance and FDC models being explained. The selected normal samples may lie far from the true decision boundary for a given fault, leading to attribution paths that pass through irrelevant regions of the input space. In contrast, ABIGX dynamically tailors the baseline for each fault sample by aligning with the FDC model's internal fault-normal decision boundary. This allows ABIGX to better isolate the minimal feature changes responsible for triggering the fault, resulting in attributions that are both more accurate and more interpretable in practice.

Moreover, ABIGX is not only visually more informative but also theoretically grounded. In Section V and Section VI, we will analyze ABIGX on a numerical fault classification and bridge our proposal with classical fault diagnosis (e.g., CP and RBC), offering a unified explanation framework for both detection and classification tasks across various model types.

4) Limitations: While ABIGX offers a general, interpretable, and theoretically grounded framework for fault explanation, it also has several limitations.

First, the method relies on iterative gradient-based optimization to compute the AFR and IG algorithms, which can be computationally intensive, especially for large-scale deep models like ViT. Although the runtime remains acceptable for most FDC scenarios (as discussed in Section VII-E), for large models such as ViT, it may become a bottleneck for real-time applications without further acceleration techniques.

Second, the "normal-like" AFR baseline should be interpreted as an approximate projection onto a model-induced normality region in the chosen representation space, instead of the normal data manifold. In particular, AFR is most reliable when (i) normal samples form a compact structure in the selected space so that the SPE objective meaningfully reflects proximity to normality, (ii) the representation mapping is locally stable and

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non-collapsed so the SPE objective remains optimizable, and (iii) the perturbation budget is sufficiently small with valid-domain projections to avoid off-support baselines. When these conditions are violated, AFR may yield less faithful baselines and thus noisier attributions, the failure cases of which are studied in Section VII-F3.

Third, though ABIGX doesn't necessarily require model internal architecture and parameters, it relies on gradient access to the underlying FDC model. This requirement makes ABIGX incompatible with models that are non-differentiable, such as tree-based models or classical Bayesian networks. While this constraint is common among gradient-based explainers, it restricts the applicability of ABIGX to only those models where backpropagation or automatic differentiation is available, which could be addressed through gradient approximation techniques in future work.

Lastly, ABIGX explains the behavior of a trained FDC model by producing associational attributions along an AFR-optimized baseline, and thus should not be interpreted as certified causal root causes. In practice, it helps narrow down candidate fault sources. If causal structure or process-mechanism priors are available, they can be incorporated as additional AFR constraints to enforce physically plausible counterfactuals.

V. ABIGX on Fault Classification

In this section, we introduce the *fault class smearing* problem, an inherent issue in fault classification models that leads to misleading variable contributions and makes it challenging to accurately identify the true cause of a fault. Aiming at this issue, we can conduct theoretical analyses on the correctness of variable contributions in fault classification. We show that the saliency map is directly affected by *fault class smearing*, while Integrated Gradients (IG) offers some alleviation. However, our analyses prove that ABIGX provides the best mitigation of this problem, delivering the most accurate explanations. These findings establish a strong theoretical foundation for ABIGX's superior explainability on fault classification models.

A. Fault Classification Setup

Firstly, we construct a toy example for the fault classification dataset and model. Without loss of generality, we assume different fault types happen on the different variables:

Example 1 (Toy fault classification dataset): Let all the normal variables independently and identically sampled from a Gaussian distribution $\mathcal{N}(0, \sigma^2 I)$, all faults have the same magnitude f and corrupted variables subject to the distribution $\mathcal{N}(f, \sigma^2 I)$. For simplicity, we assume the fault type y has one corrupted variable on the direction ξ_y. Then the fault classification dataset with M variables and N fault types is defined by $(N < M)$:

$$\begin{cases} y = 0, & x_1, x_2, \ldots, x_M \overset{i.i.d}{\sim} \mathcal{N}(0, \sigma^2) \\ y \sim \{1, 2, \ldots, N\}, & x_y \sim \mathcal{N}(f, \sigma^2),\ x_{j \neq y} \overset{i.i.d}{\sim} \mathcal{N}(0, \sigma^2) \end{cases} \tag{23}$$

Secondly, we construct the linear classification model for the toy example to best separate all the classes:

Definition 6 (Linear classifier): Let $\mathbf{W} = [\mathbf{w}_0, \mathbf{w}_1, \ldots, \mathbf{w}_N]^T$ be the weight of linear classifier, where its row vector $\mathbf{w}_y \in \mathbb{R}^{1 \times M}$ $(y \in \{0, 1, \ldots, N\})$ corresponds to the linear decision boundary of each fault (or normality) class. Based on Fisher's linear discriminant [34], the optimal row vector is trained by maximizing the distance between class mean values and minimizing the variance within classes:

$$\mathbf{w}_y^* = \arg\max \frac{\mathbf{w}_y^T (\boldsymbol{\mu}_y - \overline{\boldsymbol{\mu}})(\boldsymbol{\mu}_y - \overline{\boldsymbol{\mu}})^T \mathbf{w}_y}{\mathbf{w}_y^T \Sigma \mathbf{w}_y}$$
$$s.t.\ \|\mathbf{w}_y\|_1 = 1 \tag{24}$$

where $\boldsymbol{\mu}_y$ is the mean vector of fault class y, $\overline{\boldsymbol{\mu}}$ is the mean of class means, and $\Sigma = \sum_{y=0}^{N} \sum_{\mathbf{x} \in \mathbf{x}_y} (\mathbf{x} - \overline{\boldsymbol{\mu}})(\mathbf{x} - \overline{\boldsymbol{\mu}})^T$ is the sum of covariances per class.

Based on the dataset in Example 1 and linear classifier in Definition 6, we can prove the optimal model weights:

Theorem 2: Let $w_{y,i} \in \mathbf{W}^*$ be the optimal model weight for variable i on fault class y. For normality class $y = 0$, the optimal $w_{0,i}$ is:

$$w_{0,i} = \begin{cases} -\frac{1}{N}, & i \in \{1, \ldots, N\} \\ 0, & i \in \{N+1, \ldots, M\} \end{cases} \tag{25}$$

For fault class y, the optimal $w_{y,i}$ is:

$$w_{y,i} = \begin{cases} \frac{N}{2N-1}, & i = y \\ -\frac{1}{2N-1}, & \in \{1, \ldots, N\} \setminus y \\ 0, & i \in \{N+1, \ldots, M\} \end{cases} \tag{26}$$

Proof: The proof is provided in Appendix A11, available online. □

B. Fault Class Smearing

1) Model Weights: In this section, we formally introduce the problem of fault class smearing, an inherent challenge in fault classification models, and analyze its impact on gradient-based explanation methods. Fault class smearing arises when the classifier's optimization objective leads to non-zero weights on irrelevant variables, which in turn distorts the attribution scores of explainers such as saliency maps and integrated gradients (IG).

Definition 7 (Fault class smearing in model weights): Given the explained fault sample $\mathbf{x}$ of fault type y (defined in Example 1), and the fault classifier with optimal weights $\mathbf{W}^*$, the variable contribution are smeared into other irrelative variables in model weights:

$$w_i = \begin{cases} w_1, & i = y \\ w_2, & i \in \{1, \ldots, N\} \setminus y \end{cases} \tag{27}$$

where $w_2 \neq 0$. These non-zero weights on irrelevant fault variables cause their contributions to be overestimated by gradient-based explainers, leading to misleading attributions.

To quantitatively evaluate the extent of fault class smearing, we define the Fault Class Smearing FCS metric, which measures the proportion of total attribution incorrectly assigned to

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irrelevant variables:

$$FCS = \frac{\sum_i^{i \neq y} \text{explainer}_i}{\text{explainer}_y} \tag{28}$$

Intuitively, fault class smearing occurs because the classifier must not only separate faulty samples from normal ones but also discriminate among different fault types. This leads to shared weights across multiple classes, where variables associated with other fault types still receive non-zero weights (e.g., w_2), despite being irrelevant to the current fault class y. On the other hand, variables that do not correspond to any fault class (i.e., the remaining $M - N$ variables as shown in Eq. 44) typically have zero weights and do not contribute to smearing.

This phenomenon is problematic for model interpretation: when explanation methods rely directly on model gradients or weights (e.g., saliency maps or IG), the variable contributions may reflect class-separability bias rather than the fault-related root variables. In the following subsections, we theoretically analyze how this smearing affects different explanation methods and how ABIGX mitigates it.

2) Saliency Map: Saliency map simply calculates the absolute value of model gradient with regard to the input variables. The gradient of linear models is its weight, so that saliency map contribution is directly suffered from the fault class smearing issue.

Theorem 3 (Fault class smearing in saliency map): Given the explained fault type y with the ground-truth variable y, the saliency map contributes are smeared into other irrelative variables:

$$Grad_i = \begin{cases} |w_1|, & i = y \\ |w_2|, & i \in \{1, \ldots, N\} \setminus y \end{cases} \tag{29}$$

The degree of FCS in saliency map is:

$$FCS_{grad} = \frac{N-1}{N} \tag{30}$$

3) IG: These smearing weights on the irrelevant variables are inevitable in the fault classification models. A good explanation method should effectively mitigate the influence of smearing weights on the variable contributions. Next, we discuss fault class smearing in IG methods.

Theorem 4 (Fault class smearing in IG): Given the explained fault type y with the ground-truth variable y and the baseline of the expectation of normality samples $\mathbf{x}' = \overrightarrow{0}$, the expectation of IG contributes are:

$$\mathbb{E}_{\mathbf{x} \sim \mathbf{X}_y} IG_i = \begin{cases} |f w_1|, & i = y \\ |\frac{\sigma\sqrt{2}}{\sqrt{\pi}} w_2|, & i \in \{1, \ldots, N\} \setminus y \end{cases} \tag{31}$$

The degree of fault class smearing (FCS) in IG is:

$$FCS_{IG} = \frac{N-1}{N} \frac{\sigma\sqrt{2}}{f\sqrt{\pi}} < FCS_{grad} \tag{32}$$

Proof: The proof is provided in A12, available online. □

Theorem 4 shows the IG mitigates the fault class smearing by introducing the input variable difference between the fault and normality. Since the fault variable always has a larger offset than other variables, its integrated gradient is also more significant.

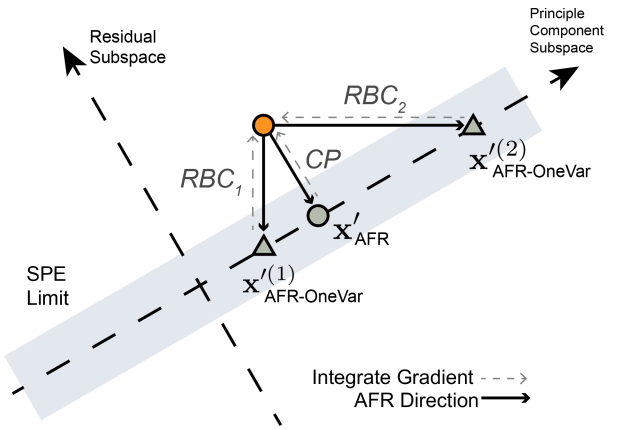


Fig. 5. Illustration of ABIGX on 2-D PCA-based detection: CP is identical to the integrated gradients between the explained sample and $\mathbf{x}'_{AFR}$; RBC is identical to the integrated gradients between the explained sample and $\mathbf{x}'_{AFR^{OV}}$ on each variable direction.

4) ABIGX: Subsequently, we analyze fault class smearing in ABIGX, showing that ABIGX outperforms IG with a lower FCS degree.

Theorem 5 (Fault class smearing in ABIGX): The fault class smearing degree in ABIGX is lower than that in IG:

$$FCS_{ABIGX} < FCS_{IG}$$

Proof: The proof is provided in Appendix A13, available online. □

Theorem 5 shows ABIGX effectively mitigates the fault class smearing problem by utilizing the gradient of classification SPE objective and outperforms IG which simply introduces the variable difference in the input space.

5) Comparison With Fault Smearing: Fault smearing [7], [35] previously discussed in CP and RBC works is caused by the same weights shared by all the inputs, since the detection models only output one scalar (e.g., SPE fault index). Though the model weights of each fault class are unique, we show that the classifiers still weigh on the irrelevant variables (due to the task for distinguish different fault types). These non-zero irrelevant weights smear the variable contributions. In summary, fault class smearing demonstrates the phenomenon and cause of the fault smearing problem in fault classification.

VI. ABIGX on Fault Diagnosis

This section presents how ABIGX uniformly reframes the two basic fault diagnosis methods, CP and RBC. For the PCA-based fault detection, we state that both CP and RBC are the specifications of ABIGX. Fig. 5 intuitively shows the identicality of CP and RBC to ABIGX with two specific AFR-reconstructed samples, $\mathbf{x}'_{AFR}$ and $\mathbf{x}'_{AFR^{OV}}$.

In the following, the PCA-based detector is denoted by its residual subspace projection matrix, $\tilde{\mathbf{C}} = \mathbf{I} - \mathbf{P}\mathbf{P}^T$.

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A. RBC and ABIGX-OneVar

Section IV discusses that ABIGX is derived from the idea of RBC. Furthermore, we prove that RBC is a linear specification of ABIGX with the AFR-OneVar.

Lemma 1 (Fault reconstruction properties [7]): The value of f_i in fault reconstruction such that $SPE(\mathbf{x} - f_i\xi_i; \tilde{\mathbf{C}})$ is minimized is:

$$f_i = (\xi_i^T \tilde{\mathbf{C}} \xi_i)^{-1} \xi_i^T \tilde{\mathbf{C}} \mathbf{x} \tag{33}$$

Theorem 6 (Identicality between RBC and ABIGX): RBC is identical to ABIGX with one-variable AFR (i.e., ABIGX-OneVar):

$$RBC_i = ABIGX_i^{OV} \tag{34}$$

Proof: The proof is provided in Appendix A21, available online. □

B. CP and ABIGX

We prove that CP is also a linear specification of ABIGX with ℓ_2 norm AFR. Firstly, we calculate the AFR-reconstructed sample for PCA under ℓ_2 norm distance:

Lemma 2 (AFR for PCA): Under ℓ_2 norm distance constraint, the AFR-reconstructed sample $\mathbf{x}'_{AFR}$ for PCA-based fault detection is:

$$\mathbf{x}'_{AFR} = \mathbf{x} - \tilde{\mathbf{C}}\mathbf{x} \tag{35}$$

which means the reconstruction vector is exactly the residual subspace projection.

Proof: The proof is provided in Appendix A22, available online. □

Subsequently, we can prove the identicality between CP and ABIGX.

Theorem 7 (Identicality between CP and ABIGX): CP is identical to ABIGX with ℓ_2 norm AFR:

$$CP_i = ABIGX_i \tag{36}$$

Proof: With $\mathbf{x}'_{AFR}$ in Lemma 2, the ABIGX is (omit the subscript of $\mathbf{x}'_{AFR}$):

$$\begin{aligned} ABIGX_i &= \xi_i^T \tilde{\mathbf{C}}\mathbf{x} \int_{\alpha=0}^{1} \frac{\partial SPE(\mathbf{x}' + \alpha\tilde{\mathbf{C}}\mathbf{x})}{\partial \mathbf{x}_i} d\alpha \\ &= \xi_i^T \tilde{\mathbf{C}}\mathbf{x} \xi_i^T (2\tilde{\mathbf{C}}\mathbf{x}' + \tilde{\mathbf{C}}\mathbf{x}) \end{aligned} \tag{37}$$

From (35), we have $\tilde{\mathbf{C}}\mathbf{x}' = \tilde{\mathbf{C}}(\mathbf{x} - \tilde{\mathbf{C}}\mathbf{x}) = 0$, thus the ABIGX contribution is obtained:

$$ABIGX_i = (\xi_i^T \tilde{\mathbf{C}}\mathbf{x})^2 \tag{38}$$

which is identical to CP_i. □

Additional, inspired by Lemma 2, we discuss the feasibility of gradient-based attack, which is the default algorithm of AFR and ABIGX:

Remark 1 (Gradient-based attack on PCA): Gradient-based attack always reconstructs fault in the direction of residual subspace projection.

Proof: The adversarial perturbation at each step of PGD is:

$$\frac{1}{L}\nabla_{\mathbf{x}} SPE(\mathbf{x}; \tilde{\mathbf{C}}) = \frac{2}{L}\tilde{\mathbf{C}}\mathbf{x} \propto \tilde{\mathbf{C}}\mathbf{x} \tag{39}$$

where L is a scalar. □

This is consistent with intuition since the residual projection points towards the direction that SPE descends fastest.

VII. Experiments

A. Evaluation Metrics and Compared Methods

We introduce two kinds of quantitative metrics for the fair evaluation of explanation methods:

Correctness [36] measures the difference between variable contribution and the root variable. The better explanation should be closer to the ground-truth. Specifically, *Correctness-AUC* treats the contributions as binary classification prediction scores. By changing the threshold of contribution scores to be negative class, the area under the receiver operating characteristic curve is calculated [37]. The correctness can also be measured by summing up the attribution scores on the ground-truth, called *Correctness-SUM*.

Consistency [38] tests whether contributions focus on where the model is truly looking. The better explanation methods should be more consistent with model behavior. The metric *Consistency-ADD* gradually adds the variable values of the explained fault sample to the normality sample. By sliding a contribution threshold, the variable with the largest contribution is first added and the least the last. The better explanation method should increase the model prediction[7] more quickly, which can be quantized by the area under the prediction score curve w.r.t threshold. Conversely, *Consistency-DEL* first deletes the most important variables until all the features are replaced by the normality sample. Similarly, the better method should decrease the model prediction faster, which can be measured by the AUC w.r.t. threshold.

For the baselines, we introduce several advanced gradient-based XAI methods, including:

- *Saliency map:* Absolute value of input variable gradient, the most common feature attribution method.
- *DeepLift [39]:* A gradient backpropagation explanation method that can properly handle cases where IG may give misleading results.
- *Vanilla IG [16]:* The original IG method proposed, using the all-zero vector as the baseline.
- *Expected IG [40]:* An IG variant using the informative baselines (normal samples in our experiments) from the data distribution, mitigating the impact of baseline choice by the expectation of IG attributions.
- *Important Direction Gradient Integration (IDGI) [41]:* An IG variant directly focusing solely on the important direction that aligns with the gradient, built on the vanilla IG in the experiment.

[7]The model prediction for classification is fault type confidence with Softmax function, and that for detection is SPE value. When calculating the AUC of detection models, we normalize all SPE values to the range [0,1].

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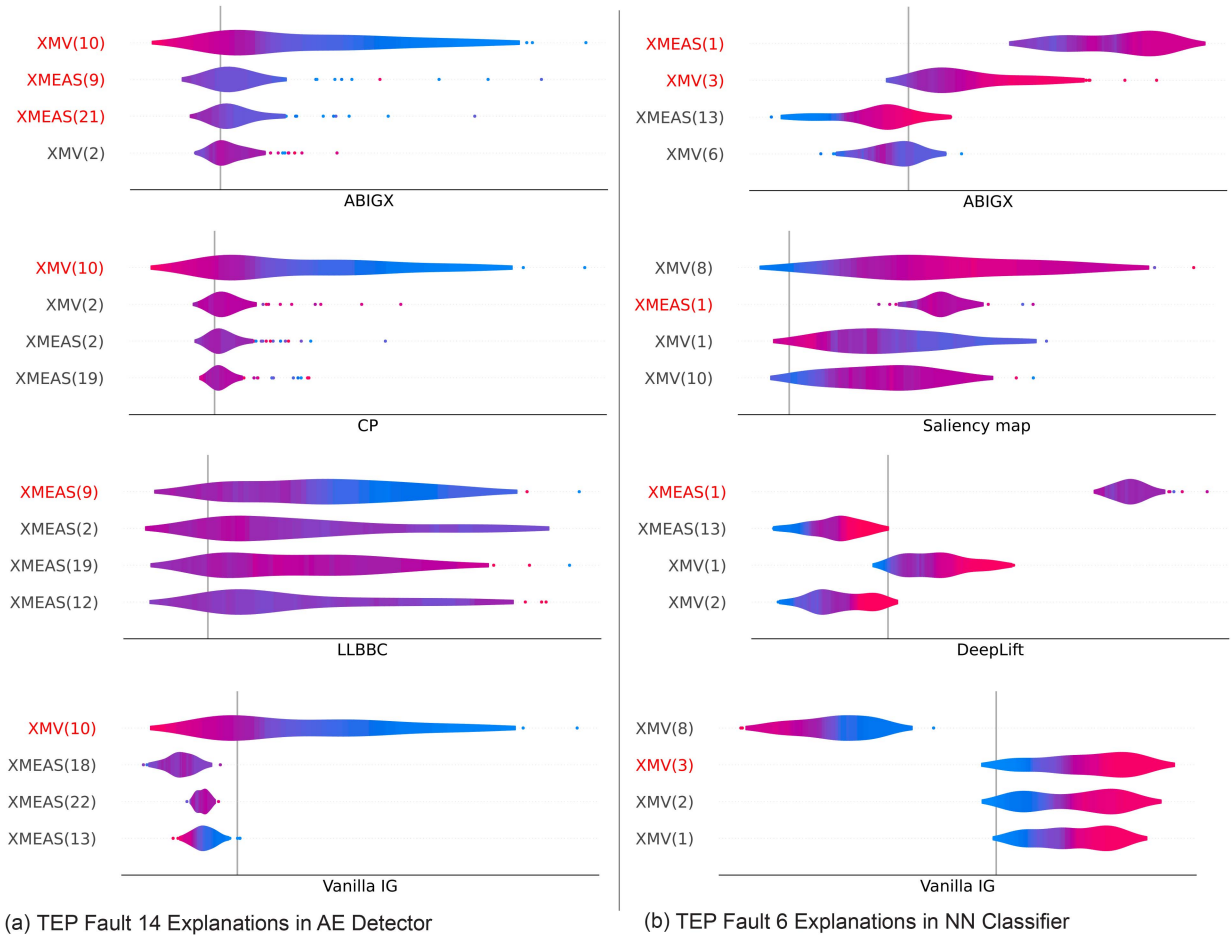


Fig. 6. Explanations comparison for the NN classifier on TEP Fault 6, where the ground-truth root variables are in red. Fault 6 is caused by root variables XMEAS(1) and XMV(3) and Fault 14 is caused by XMEAS(9), XMV(10), and XMEAS(21), detailed in Table B.2.

- *Guided IG [37]:* An IG variant constructs the path based on the gradient, using the all-zero vector as the baseline.
- *AFR + Guided IG (ablation settting):* we use the AFR-reconstructed sample as the baseline, but replace ABIGX's default straight-line integration with the path construction algorithm of Guided IG, used as an ablation setting to verify the effectiveness of AFR baseline in ABIGX.

To ensure a fair comparison, we set the primary hyperparameter, step size, for all IG-based methods (Vanilla IG, Expected IG, Guided IG and IDGI) equal to that used in ABIGX. All the experiments are conducted on an INTEL CORE i7-12700 K CPU for PCA-, AE-, and MLP-based models on TEP, and on a single NVIDIA GeForce RTX 3080 GPU for CNN on Wafermap and ViT on MVTec-AD.

B. Tennessee Eastman Process (TEP)

1) Dataset Description: To make the TEP setting self-contained and to clarify what each fault means physically, we provide the TE process flowchart and the official variable and fault descriptions from [18] in the Appendix (Fig. B.1, Table B.1), available online. These materials specify the fault injection locations (streams, units), the actuators and sensors involved, and the expected first-order responses, which are crucial for interpreting both diagnosis performance and feature-level explanations.

Our explanation-correctness metrics rely on a curated set of root variables for each fault, which are derived from canonical TE process documentation and summarized in Appendix Table B.2, available online. Beyond listing variable indices, the table provides a fault-by-fault rationale linking each fault to its root variables based on (i) injection location, (ii) the most directly affected measurements and actuators, and (iii) the process structure and control loops. We use it as the ground-truth reference when reporting root-variable-based correctness scores. This fault-to-variable linkage is also consistent with the fault semantics adopted in representative TEP diagnosis works [9], [42], [43], making our fault-to-variable mapping a transparent, literature-aligned representation of canonical TEP fault definitions for explanation evaluation.

For clarity, Fig. 6 intuitively display the explanation comparisons for two TEP fault types, Fault 6 and Fault 14. The figure respectively display sensor variable contributions of fault detection and classification models, where the four subplots of each figure show the variable contributions explained by four different techniques. For TEP dataset, we

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TABLE II
MEAN DIFFERENCE BETWEEN VARIABLE CONTRIBUTION OF CP, RBC, AND ABIGX (ON PCA DETECTOR FOR TEP)

AFR	ABIGX		ABIGX-OneVer	
	PGD	MILP	PGD	MILP
CP	1.62×10^{-6}	1.25×10^{-6}	-	-
RBC	-	-	9.83×10^{-4}	1.15×10^{-5}

TABLE III
EVALUATION METRICS OF EXPLANATION METHODS (ON AE DETECTOR FOR TEP)

Metrics	Correctness		Consistency	
	AUC↑	SUM↑	ADD↑	DEL↓
ABIGX	0.382	0.041	**0.922**	**0.256**
ABIGX-OneVar	**0.478**	**0.055**	0.903	0.275
CP	0.326	0.028	0.896	0.283
LLBBC	0.314	0.029	0.838	0.330
Saliency map	0.470	0.051	0.897	0.292
Vanilla IG	0.381	0.035	0.903	0.274
Expected IG	0.395	0.045	0.913	0.267
IDGI	0.389	0.040	0.889	0.290
Guided IG	0.390	0.044	0.900	0.281
AFR+Guided IG	0.412	0.048	0.925	0.249

use the plot style from the work [44]. The horizontal axis represents the variable contribution value (with direction). The variables are sorted by importance on the vertical axis, where the four variables with the most contributions are displayed. Notably, the plots also show the correlation between variable (feature) value and its contribution by the colored distribution.

2) Explainable Fault Detection. Identicality between Fault Diagnosis and ABIGX: Section VI theoretically proves the identicality between two fault diagnosis methods and ABIGX. On the PCA detector for TEP faults, we compute the empirical error between them, which is the mean value over contributions of all variables. Table II reports the results, which shows the differences are small enough (especially with AFR-reconstructed samples via MILP).

Explanation results: Firstly, Fig. 6(a) plots the intuitive comparisons between the different explanations on AE models for the TEP dataset, showing that ABIGX assigns the highest contributions to the root variables. At the same time, other methods do not provide complete explanations.

Table III reports four evaluation metrics on the explanations of AE detection for TEP, where 14 Fault types are concerned. Overall, ABIGX methods outperform other explanations, including both fault diagnosis methods (CP and LLBBC [9]) and IG-based deep learning explainers (DeepLift is not adapted for fault detection). By comparing Expected IG with Vanilla IG and IDGI, we can find that the all-zero baseline has adverse effect on the fault detection explanation. ABIGX-OneVar provides the explanations that are closest to the ground-truth root variables. The saliency map that attributes variables independently also performs well on the correctness metrics. On the other hand, ABIGX provides explanations that best match the model behavior. We argue that the explanations concerning variable interaction are closer to how models actually deal with the samples.

TABLE IV
EVALUATION METRICS OF EXPLANATION METHODS (ON NN CLASSIFIER FOR TEP)

Metrics	Correctness		Consistency	
	AUC↑	SUM↑	ADD↑	DEL↓
ABIGX	0.557	**0.075**	**0.890**	**0.032**
ABIGX-OneVar	0.584	0.073	0.723	0.121
ABIGX-AdvAFR	**0.608**	0.074	0.726	0.135
Saliency map	0.573	0.072	0.732	0.133
DeepLift	0.400	0.037	0.876	0.042
Vanilla IG	0.313	0.024	0.726	0.083
Expected IG	0.415	0.045	0.886	0.043
IDGI	0.370	0.031	0.821	0.052
Guided IG	0.401	0.040	0.875	0.039
AFR+Guided IG	0.515	0.056	0.882	0.036

Based on the ablation study (AFR+Guided IG), using the AFR-reconstructed baseline together with Guided IG's adaptive path consistently improves vanilla Guided IG on TEP dataset, while achieving competitive performance relative to ABIGX.

3) Explainable Fault Classification: Similar to fault detection, Fig. 6(b) plots the explanation of NN-based fault classifier on TEP Fault 6. The advantage of ABIGX is on Variable XMV(3). ABIGX assigns higher contributions to this variable, while the saliency map even ignores it, the reason for which has been analyzed in Section IV-C2.

Table IV reports the explanation metrics on fault classification, where the method performances are similar to the detection. ABIGX still generally performs better, even under the correctness (except AUC). ABIGX-OneVar still emphasizes correctness but has poor model consistency, which is similar to the saliency map. DeepLift and vanilla IG have similar performances. Though IDGI and Guided IG show performance improvement over original IG method, they are not competitive in the fault classification even with expected due to their non-imformative baseline.

Validation of classification SPE index: An ablation study is performed with ABIGX-AdvAFR. Instead of ABIGX's classification SPE proposed in Section III-B, ABIGX-AdvAFR reconstructs faults by the original adversarial attack objective (i.e., minimizing Softmax output confidences of normal class). The overall better performance of ABIGX than ABIGX-AdvAFR validates the effectiveness of the novel classification SPE index for fault reconstruction.

4) Fault-Level Explanations: To provide a clear picture of attribution comparison on TEP dataset, Appendix Fig. B.2 and

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B.3 (available online), along with Table B.4 and B.3, present detailed experimental results across all TEP fault types for both fault detection and classification models. The figures visualize ABIGX's attributions for top-ranked variables across the data distribution, while the tables compare attribution performance across methods by reporting root variable rankings at the fault level. Specifically, the ratios of root variables appearing in the top-3 attributed variables, along with their average rankings, show that ABIGX consistently achieves the highest alignment with ground-truth fault variables across most fault types.

To explore the correlation between model accuracy and ABIGX explanation quality, Fig. B.4 presents a detailed analysis of ABIGX's attribution performance (Correctness-SUM) across all TEP fault types, separated into correctly and incorrectly predicted samples for different FDC models. The results highlight two key observations: (1) fault types with higher prediction accuracy tend to exhibit more accurate ABIGX attributions; and (2) within each fault type, correctly predicted samples generally yield higher attribution correctness than misclassified ones. These findings indicate that ABIGX explanations are closely aligned with the model's confidence, which faithfully attributes meaningful contributions when the model detects faults successfully, and revealing attribution uncertainty when the model fails.

C. Wafer Map

1) Dataset Description: Wafer map analysis is critical in daily semiconductor manufacturing operations. Wafer maps provide visual details that are crucial for identifying the stage of manufacturing at which wafer fault pattern occurs. We study the explanation of the wafer map fault pattern classification, the target of which is to determine which pixels of the wafer maps are the fault root.

We train a convolution neural network (CNN)-based fault classifier on WM-811 K dataset [45], of which four fault types (Center, Edge-Loc, Edge-Ring, and Scratch) are selected. The fault and normal wafer maps are displayed in Fig. 7, where the ground-truth root pixels are circled by red lines. The green pixels denote the normal dies, and the yellow pixels are the defective dies. A wafer map is normal when there is no certain defective die pattern (e.g., the top left one in Fig. 7).

2) Explainable Fault Classification: The experimental results presented in Fig. 7 and Table V provide a comparative analysis of various explanation methods for a CNN classifier used to identify fault patterns in wafer maps. ABIGX stands out for producing highly focused and sharp explanations across all fault types, closely aligning with the highlighted ground-truth areas. This clarity suggests that ABIGX effectively identifies the regions most relevant to the model's classification decisions. Other methods, such as Saliency Map, Vanilla IG and IDGI, produce more scattered and less focused explanations. We can find that the the black baseline used in vanilla IG and IDGI causes poor explanation. DeepLift and Expected IG perform moderately well, offering clearer explanations than Saliency Map and Vanilla IG but lacking the precision seen in ABIGX.

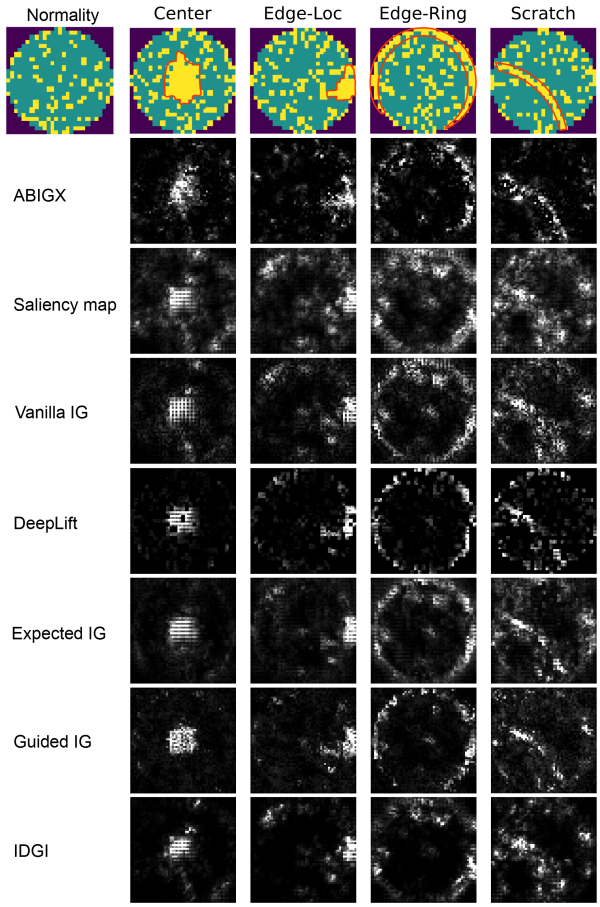


Fig. 7. Explanations for CNN classifier on four sampled wafer maps with different fault patterns, where the ground-truth root pixels are circled in red.

TABLE V
EVALUATION METRICS OF EXPLANATION METHODS (ON CNN CLASSIFIER FOR WAFER MAPS)

	Correctness		Consistency	
Metrics	AUC↑	SUM↑	ADD↑	DEL↓
ABIGX	**0.839**	0.524	**0.859**	**0.040**
Saliency map	0.571	0.298	0.623	0.236
DeepLift	0.812	**0.541**	0.806	0.053
Vanilla IG	0.706	0.381	0.699	0.111
Expected IG	0.814	0.477	0.796	0.052
IDGI	0.760	0.432	0.742	0.068
Guided IG	0.820	0.499	0.813	0.047
AFR+Guided IG	0.844	0.531	0.847	0.040

Table V quantifies the performance of each method across four key metrics. ABIGX achieves the highest scores in both correctness (AUC: 0.839) and consistency (ADD: 0.859, DEL: 0.040), indicating its superior performance in accurately identifying critical regions while remaining consistent in the explanations generated. In comparison, Saliency Map and Vanilla IG generally exhibit the lowest performance. DeepLift and Expected IG perform well in terms of correctness, but they

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TABLE VI
EVALUATION METRICS OF EXPLANATION METHODS (ON VIT ANOMALY DETECTOR FOR MVTEC-AD DATASET)

	Correctness		Consistency	
Metrics	AUC↑	SUM↑	ADD↑	DEL↓
ABIGX	**0.918**	**0.361**	0.997	0.490
Attention map	0.913	0.173	0.992	0.535
Saliency map	0.809	0.215	0.992	0.478
Vanilla IG	0.878	0.181	0.995	0.508
Expected IG	0.901	0.224	0.995	0.599
IDGI	0.886	0.329	**0.998**	0.523
Guided IG	0.811	0.242	0.994	**0.488**
AFR+Guided IG	0.915	0.298	0.997	0.492

fall short in consistency relative to ABIGX. Guided IG shows a balance between correctness and consistency, with scores that are competitive with but not surpassing ABIGX. Using the AFR-reconstructed baseline with Guided IG's adaptive path (AFR+Guided IG) consistently outperforms vanilla Guided IG and is competitive with ABIGX, confirming that AFR baseline improvements persist under different path choices.

The experiments on wafer map fault explanation validate the generalization of ABIGX, which is not only applicable for tabular sensor data but also for the unstructured images trained by CNN models.

3) Detailed ABIGX Explanations: Appendix Fig. B.5 (available online) presents ABIGX explanation results on more wafer maps for seven fault types. Each row shows four representative samples per fault, with the left image depicting the original wafer map and the right showing the ABIGX attribution heatmap. The Consistency-ADD score above each pair quantifies how consistent the explanations are across samples. ABIGX achieves high consistency for faults with distinct spatial patterns such as "Center," "Edge-Ring," and "Random," indicating its effectiveness in highlighting model-relevant regions aligned with visual fault characteristics.

However, ABIGX shows limitations on more complex or ambiguous faults. For instance, "Near-full" consistently yields low Consistency-ADD scores (0.375), suggesting that the model only confidently predicts this fault type when a large portion of the wafer is affected. Fault types like "Scratch," "Loc," and "Edge-Loc" also exhibit inconsistent explanation quality, highlighting ABIGX's challenges in handling mixed or irregular patterns. These observations point to the need for future refinement to improve robustness and generalizability in real-world quality inspection tasks.

D. MVTec-AD Industrial Image

1) Dataset Description: The MVTec-AD (MVTec Anomaly Detection) dataset [46] is a widely used benchmark for anomaly detection in industrial visual inspection. It contains high-resolution images of 15 object and texture categories, each with normal and defective samples exhibiting various types of surface anomalies such as scratches, dents, and color deviations. Unlike the tabular datasets used in traditional fault detection (e.g., TEP), MVTec-AD focuses on fine-grained spatial defects, making it suitable for evaluating vision-based anomaly detection models. Its diverse defect types and realistic imaging conditions pose unique challenges for interpretability, providing a valuable testbed to assess the ABIGX's generalizability.

2) Model Description: In our work, we adopt WinClip [32] as a representative ViT-based anomaly detector to evaluate the generalization ability of ABIGX. WinClip is a state-of-the-art vision transformer model designed for zero-shot anomaly detection in industrial images. By leveraging CLIP [47] embeddings and multimodal alignment, it achieves strong performance across diverse categories in the MVTec-AD dataset. We use the pretrained ViT backbone (ViT-B-16-plus-240) and make minor modifications to the WinClip implementation to ensure differentiability, enabling compatibility with gradient-based explanation methods. This experimental setting demonstrates the applicability of ABIGX to more complex, large-scale, and modern transformer-based models.

3) Explainable Anomaly Detection: Fig. 8 compares the ViT attribution maps of different methods on input industrial images from five categories (Carpet, Wood, Leather, Tile, and Grid). ABIGX delivers significantly more focused and accurate attributions compared to other baseline methods, which consistently highlights the true anomaly regions, as indicated by the red circles, while suppressing irrelevant noise. In contrast, alternative methods like the attention map and Vanilla IG exhibit dispersed or less localized attributions, often extending beyond the ground-truth faulty areas.

Quantitatively, Table VI validates the superior explanation performance of ABIGX. In terms of correctness, ABIGX achieves the highest AUC and SUM, indicating that its saliency regions align more closely with the actual fault areas compared to other approaches. Notably, this demonstrates that ABIGX's attribution maps are not only visually interpretable but also quantitatively meaningful. The performance gain over Expected IG and Saliency map further validates the advantage of ABIGX's adaptive baseline strategy, especially in complex, high-dimensional image domains where conventional IG methods struggle with arbitrary baselines.

In terms of consistency, ABIGX demonstrates strong robustness across evaluation metrics. Notably, WinClip's predictions are highly sensitive to minor perturbations, adding just a few pixels can significantly increase the anomaly score, resulting in uniformly high ADD scores across all methods. However, ABIGX achieves a relatively low DEL score, indicating that the top-ranked features it identifies are critical to the model's predictions.

Overall, this experiment confirms that ABIGX is compatible with large-scale, transformer-based models like WinClip, and produces faithful and concise explanations across diverse anomaly types in industrial imagery. The demonstrated generalizability of ABIGX to a different modality and model architecture reinforces its value as a broadly applicable and unified explainability framework for both tabular and visual FDC tasks.

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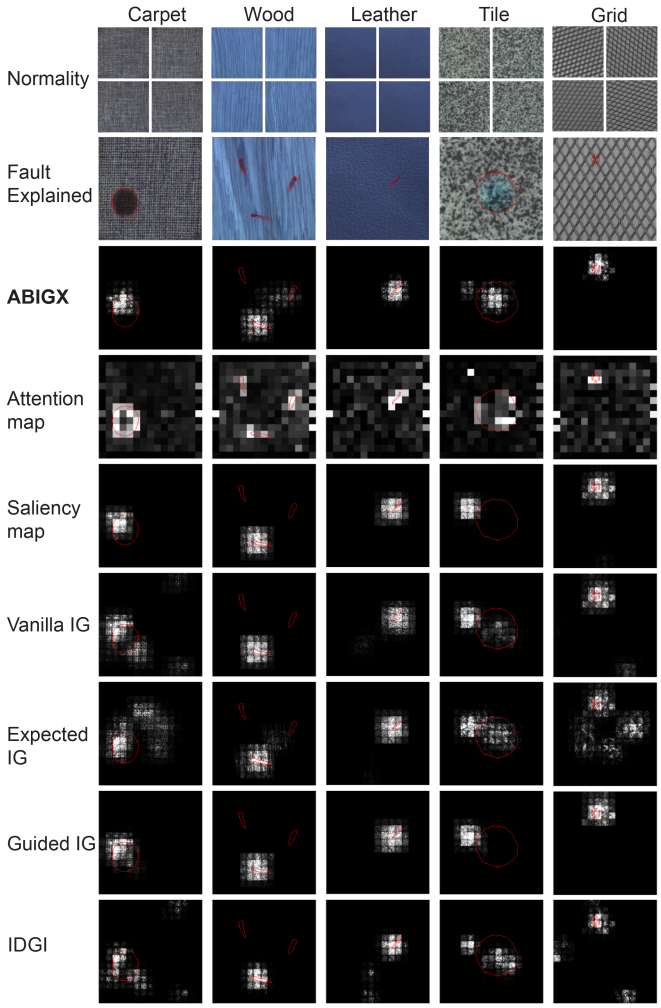


Fig. 8. Explanation results for the WinClip anomaly detector on five samples from the MVTec-AD dataset. Ground-truth anomaly regions are circled in red (zoom in to view finer details in small regions).

4) Detailed ABIGX Explanations: Appendix Fig. B.6 (available online) presents ABIGX explanations for more MVTec-AD fault samples with subtle and visually non-obvious defects. These challenging cases are meaningfully for evaluating the necessity of explanation methods. For the samples that FDC model assigns high fault confidence, ABIGX highlights localized regions that closely align with the true defect areas, suggesting that the model is attending to semantically relevant features. In contrast, the samples with lower fault confidence, where ABIGX produces weaker or misaligned attributions, reflecting the model's uncertainty or failure to recognize the anomaly. This analysis shows that (1) for defects that are hard to observe, explanations are vital for verifying that the model is making decisions based on meaningful evidence, and (2) explanations serve as diagnostic tools, helping to validate correct decisions and interpret failure cases for further model diagnosis.

E. Computational Cost

Appendix Table B.15 (available online) presents the average explanation time per sample. Despite the increased computational overhead, ABIGX remains practically efficient, with runtimes averaging around one to two seconds for lightweight models (e.g., PCA, AE, MLP, and small CNNs), and staying within tens of seconds even for more complex architectures such as ViT.

Traditional methods such as CP and RBC are computationally lightweight, with explanation times near zero for PCA models. This efficiency is due to the non-reliance on iterative optimization or gradient computation. However, this speed comes at the cost of generalizability and interpretability in nonlinear settings.

The primary computational cost in ABIGX arises from gradient computations, which are essential for both the AFR process and the IG calculation. Consequently, the total runtime is largely determined by the number of forward and backward passes through the model, governed by the step parameters used in both AFR and IG. In practice, we typically set the number of steps in AFR equal to that in IG, resulting in ABIGX requiring approximately twice the number of gradient evaluations compared to the standard IG method, and roughly $2\times$ step times more than simple gradient-based methods.

In practice, ABIGX can be substantially accelerated via mini-batching, where AFR baseline search and IG integration are executed for multiple samples in parallel under a GPU-memory-limited batch size (e.g., 8 for ViT in our setup).

F. Implementation Details

We summarize the model architectures and key ABIGX hyperparameters in the Appendix for reproducibility. Specifically, Table B.5 and B.14 reports our trained network structures in experiments; Table B.6 lists the main ABIGX hyperparameters across all four explained models.

1) Step Size, Number and Convergence: To avoid under-optimized AFR baselines affecting the attribution results, we use a conservative (large) iteration budget for all experiments with reasonable running time, so that insufficient optimization does not confound the attribution quality. The sensitive analysis of step size and number for all four explained models is presented in Appendix Table B.7 to B.10, available online.

To diagnose convergence during AFR baseline search, we monitor the representation SPE value and model prediction confidence over iterations. Specifically, we mark a run as converged when the SPE curve reaches a plateau, i.e., the relative SPE decrease becomes negligible over successive iterations. According to the aggregated convergence rates in Table B.16, under the default step size and step number, most samples exhibit a sharp SPE drop followed by saturation, indicating stable convergence behavior across models. We also track model normality during the search (Fig. B.9), which generally stabilize consistently with the SPE-based convergence diagnosis. Only a small fraction of runs remain locally trapped (without high enough normality confidence) or hit the maximum-iteration budget.

2) Fault Classifier Representation Layers: In our fault-classification setting, ABIGX requires selecting an internal representation $h(x)$ to define the classification-SPE and drive AFR baseline search. We choose the penultimate layer as the default representation, since it provides a stable, semantically meaningful feature space while avoiding the saturation and label-specific collapse that can occur at the final logit layer. We

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verify this choice via a sensitivity study on both TEP and Wafer Map classifiers: across candidate layers, the penultimate-layer representation consistently yields the best or near-best correctness and consistency of explanations (Appendix Fig. B.7 and Table B.13, available online). Therefore, we use the penultimate layer for all classifier experiments unless otherwise specified.

3) Failure Cases: We observe two recurrent failure modes of ABIGX. The first case is AFR baseline optimization failure (low normality at the final baseline). In some samples, the PGD-based AFR search reduces the representation-space distance but still converges to a baseline with relatively low model normality confidence, suggesting the optimization is trapped in a poor local optimum. When the final baseline remains non-normal, the resulting IG path no longer represents a meaningful "fault-to-normal" transition, and the attribution correctness (e.g., GT-AUC) degrades accordingly (Fig. B.8). As a practical safeguard, we can fall back to Expected IG by drawing baselines from an input-space normal pool instead of relying on the searched AFR baseline, which improves robustness in these cases.

The second case is low-confidence model predictions (intrinsically ambiguous samples).This group of failures arises when the model itself assigns low fault confidence to the explicand. In such cases, attribution may not align well with human ground truth, not because the explainer is incorrect, but because the underlying decision is uncertain (Fig. B.6). This behavior is expected: ABIGX is designed to explain the model's decision mechanism, and when the model is weakly confident (or near the decision boundary), any faithful explanation will reflect that ambiguity rather than forcing alignment with external annotations.

VIII. Conclusion

This work proposes a unified framework, ABIGX, for explaining the general FDC models. ABIGX is derived from the essentials of CP and RBC, which have succeeded in fault diagnosis of PCA models. The novel proposed fault reconstruction method, AFR, is the core part of ABIGX. AFR reframes existing methods from the adversarial attack perspective and generalizes FR to the fault classification by the novel classification SPE index. For explainable fault classification, we analyze the performances with regard to fault class smearing. Compared with the saliency map and IG, we prove that ABIGX provides more accurate variable contributions with a lower degree of fault class smearing. For explainable fault detection, we theoretically prove the identicality of linear CP and RBC to the ABIGX. With quantitative metrics, the experiment on a wide range of models and datasets validates the general superiority of ABIGX to other advanced explainers in both fields of fault diagnosis and deep learning explanation.

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